

1 (a) (i) V units: m^3 (*allow metres cubed or cubic metres*) A1 [1]

(ii) Pressure units: $\text{kg m s}^{-2} / \text{m}^2$ (*allow use of $P = \rho gh$*) M1
 Units: $\text{kg m}^{-1} \text{s}^{-2}$ A0 [1]

(b) V / t units: $\text{m}^3 \text{s}^{-1}$ B1
 Clear substitution of units for P , r^4 and l M1

$$C = \frac{\pi P r^4}{8 V t^{-1} l} = \frac{\text{kg m}^{-1} \text{s}^{-2} \text{m}^4}{\text{m}^3 \text{s}^{-1} \text{m}}$$

Units: $\text{kg m}^{-1} \text{s}^{-1}$ A1 [3]
(8 or π in final answer –1. Use of dimensions max 2/3)

2 (a) (i) $v = u + at$ C1